SECTION – A

GENERAL APTITUDE

Choose the most appropriate option.

(Q.No. 01 to 13)

1. Who has created history by becoming the first Indian to win a medal in IAAF Continental Cup?

(A) Arpinder Singh

(B) Rahi Sarnobat

(C) Neeraj Chopra

(D) Heena Sindhu

2. Abu alone can do a piece of work in 6 days and Salem alone in 8 days. Abu and Salem undertook to do it for ₹ 3,200. With the help of Ahmad , they completed the work in 3 days. How much is to be paid to Ahmad?

(A) ₹ 300

(B) ₹ 400

(C) ₹ 350

(D) ₹ 500

3. A can do a piece of work together for 10 days rest of the work is finished by C in 4 days. If C got ₹ 1,000 for complete work, what are the daily wages of A and B respectively?

(A) 300, 200

(B) 200, 300

(C) 300, 250

(D) 200, 250

4. Which Indian software company has linked to deal with Australian Open as official digital innovation partner?

(A) TCS

(B) Infosys

(C) Wipro

(D) HCL

5. Five farmers A, B, C, D, and E have 7, 9, 11, 13, and 14 mango trees, respectively in their orchards. Last year, each of them discovered that every tree in their own orchard bore exactly the same number of mangoes. Further, if the third farmer C gives one mango to The A, and the E gives three mangoes to each of the B and the D, then they would all have exactly the same number of mangoes. What were the yields per tree in the orchards of the C and D farmers?

(A) 11 and 9 mangoes per tree respectively

(B) 11 and 8 mangoes per tree respectively

(C) 8 and 11 mangoes per tree respectively

(D) 8 and 10 mangoes per tree respectively

6. Look at this series : 5.2, 4.9, 4.6, 4.3, ….. What number should come next?

(A) 4

(B) 4.1

(C) 4.2

(D) 5.1

7. Look the series 41, 47, 45, 51, 49, 55, 53, …and find the suitable next number.

(A) 50

(B) 59

(C) 49

(D) 43

8. When Rahul solve the problem then in intermediate step a number x is required to divide by 2/3 but by mistake he multiply by 3/2, then calculate percentage of error in his result.

(A) 1 %

(B) 2 %

(C) 3 %

(D) 0 %

9. Which word does NOT belong with the others?

(A) couch

(B) rug

(C) table

(D) chair

10. Today is Tuesday. After 62 days, it will be :

(A) Wednesday

(B) Thursday

(C) Monday

(D) Friday

11. According to some language, if daftafoni means misadvise, and imolokti means misconduct, then which word could mean “statement”?

(A) kratafoni

(B) kratadafta

(C) loktifoni

(D) daftaimo

12. A certain number of two digits is three times the sum of its digits. If 45 is added to it, the digits are reversed. The number is ____.

(A) 16

(B) 72

(C) 63

(D) 27

13. Look at this series : A3, D12, G21, _______,M39, P48, S57, and find the suitable number to fill in the blank.

(A) I30

(B) I31

(C) J30

(D) J31

SECTION – B

ENGINEERING MATHEMATICS

Choose the most appropriate option.

(Q.N0. 14 TO 30)

14. If f(x) = $\begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}$ then f(2x) – f(x) equals to :

(A) a(2a + 3x)

(B) ax(3x + 2a)

(C) $ax^2(2a + 3x)$

(D) ax(2a + 3x)

15. The distance between parallel lines y = 2x + 4 and 6x = 3y + 5 :

(A) $\frac{17}{\sqrt{3}}$

(B) 1

(C) $\frac{17\sqrt{5}}{15}$

(D) $\frac{3}{\sqrt{5}}$

16. Equation of straight line through (3, 2) which makes an angle 45^0 with the line x – 2y – 3 = 0.

(A) 3x – y = 7, x + 3y = 9

(B) 3x + y = 7, x – 3y = 9

(C) x – 3y = 7, x + 3y = 7

(D) 3x + y = 9, x – 3y = 7

17. A coin is tossed three times. Let X denotes number of times a head appears. Find the mean of X.

(A) 1.5

(B) 1

(C) 2

(D) 0

18. Consider the improper integral $I_1 = \int_1^{\infty} \frac{\sqrt{x}}{(1+x)^2}$ dx and $I_2 = \int_0^{-1} \frac{1}{x^2(1+x)}$dx then :

(A) I_1 converges and I_2 diverges.

(B) I_2 converges and I_1 diverges.

(C) I_1 converges and I_2 converges.

(D) I_1 diverges and I_2 diverges.

19. The rank of the matrix $\begin{matrix} 1 & 2 & 3 & 4 \\ 2 & -1 & -4 & 3 \\ 3 & 1 & -1 & 7 \\ 4 & 3 & 2 & 11 \end{matrix}$ is :

(A) 4

(B) 3

(C) 1

(D) 2

20. The mean of 200 items was 50. Later on it was discovered that two items were misread as 92 and 8 instead of 192 and 88. Find out the corrected mean.

(A) 50.9

(B) 49.9

(C) 52

(D) 51.9

21. Find Fourier series of |x| on [-π, π] is:

(A) $\frac{\pi}{2} - \frac{4}{\pi}\left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \cdots\right]$

(B) $\frac{\pi}{2} + \frac{4}{\pi}\left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \cdots\right]$

(C) $\frac{\pi}{2} - \frac{4}{\pi}\left[\frac{\sin x}{1^2} + \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} + \cdots\right]$

(D) $\frac{\pi}{2} + \frac{4}{\pi}\left[\frac{\sin x}{1^2} + \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} + \cdots\right]$

22. The particular integral of $[D^2+1]y = e^x \sin x$ is:

(A) $\frac{e^x}{5}$ (2sinx + cosx)

(B) $-\frac{e^x}{5}$ (2cosx – sinx)

(C) $\frac{e^x}{5}$ (cosx – 2sinx)

(D) $-\frac{e^x}{5}$ (2cosx – 3sinx)

23. Evaluate $\iint \sin\pi(x^2+y)\,dxdy$ over interior of circle $x^2 + y^2 = 1$

(A) -2

(B) 1

(C) 2

(D) -1

24. The partial differential equation formed by eliminating arbitrary functions f & g from

z = f (x + at) + g (x – at) is:

(A) $\frac{\partial^2 z}{\partial t^2} - a^2 \frac{\partial^2 z}{\partial x^2} = 0$

(B) $a \frac{\partial^2 z}{\partial t^2} + \frac{\partial^2 z}{\partial x^2} = 0$

(C) $\frac{\partial^2 z}{\partial t^2} + a^2 \frac{\partial^2 z}{\partial x^2} = 0$

(D) $a^2 \frac{\partial z}{\partial t} + \frac{\partial z}{\partial x} = 0$

25. Which of the following statement is true?

(A) The function f (x, y) at a point increases most rapidly in the direction of – *gradf* at that point.

(B) The function f (x, y) at a point decreases most rapidly in the direction of *gradf* at that point.

(C) The function f (x, y) at a point has no change in the direction parallel to direction of *gradf* at that point.

(D) The function f (x, y) at a point has no change in the direction perpendicular to direction of *gradf* at that point.

26. Using Rolle's theorem the equation:

$a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots\ldots + a_n = 0$ has at least one root between o and 1 if :

(A) $\frac{a_0}{n} + \frac{a_1}{n-1} + \ldots\ldots + a_{n-1} = 0$

(B) $\frac{a_0}{n-1} + \frac{a_1}{n-2} + \ldots . + a_{n-2} = 0$

(C) $na_o + (n-1)a_1 + \ldots . + a_{n-1} = 0$

(D) $\frac{a_0}{n+1} + \frac{a_1}{n} + \ldots . + a_n = 0$

27. The solution of $\frac{dy}{dx} = \log\left[x\frac{dy}{dx} - y\right]$ is:

(A) $y = cx - e^c$

(B) $c = \log(-cx + y)$

(C) $x = cy + e^c$

(D) $c = \log(cy + x)$

28. The determinant $\begin{vmatrix} xp + y & x & y \\ yp + z & y & z \\ 0 & xp + y & yp + z \end{vmatrix} = 0$ if:

(A) x, y, z are in Arithmetic progression.

(B) x, y, z are in Geometric progression.

(C) x, y, z are in Harmonic progression.

(D) xy, yz, zx are in Arithmetic progression.

29. Let X is a random variable with binomial distribution having parameters n and p. If mean and variance of X are 5 and 4 respectively. Find n + p + q is:

(A) 24

(B) 25

(C) 26

(D) 27

30. If A = $\begin{pmatrix} cosx & sinx \\ -sinx & cosx \end{pmatrix}$ and A (adjA) = k $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ then value of k is :

(A) sinx cosx

(B) 1

(C) 2

(D) 3

SECTION – C

COMPUTER SCIENCE

Choose the most appropriate option.

(Q.No. 31 to 85)

31. Which register that shift complete binary number in one bit at a time and shift all the stored bits out at a time?

(A) Parallel-in parallel-out
(B) Parallel-in Serial-out
(C) Serial-in parallel-out
(D) Serial-in Serial-out

32. Which among the following algorithm can't be used with linked list?

(A) Binary search
(B) Linear Search
(C) Insertion sort
(D) Merge Sort

33. A program P reads in 1000 integers in the range [0, 100] representing the scores of 500 students. It then prints the frequency of each score above 50. What would be the best way for P to store the frequencies?
(A) An array of 50 numbers
(B) An array of 100 numbers
(C) An array of 500 numbers
(D) A dynamically allocated array of 550 numbers

34. In a risk based approach the risks identified mayn't be used to:
(A) Determine the test technique to be employed
(B) Determine the extent of testing to be carried out
(C) Prioritize testing in an attempt to find critical defects as early as possible
(D) Determine the cost of the project

35. Simple network management protocol(SNMP) is a protocol that runs on:
(A) Application layer
(B) Transport layer
(C) Network layer
(D) Data link layer

36. In a database system, the domain integrity is not defined by:
(A) The data type and the length
(B) The NULL value rejection
(C) The allowable values, through techniques like constraints or rules
(D) Default value

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37. Transformation that moves objects without deformation is called:
(A) Rotation
(B) Scaling
(C) Translation
(D) Morphism

38. Which of the following need not necessarily be saved on a context switch between processes?
(A) Program counter
(B) IO status information
(C) CPU registers
(D) Translation lookaside buffer

39. Given an random unsorted array ‘A’ in which every element is at most ‘d’ distance from is position in sorted array where d<Size(A). If we applied the insertion sort over this array, then the time complexity of algorithm is:

(A) O(nlogd)
(B) $O(n^2 \log d)$
(C) O(nd)
(D) $O(n^2 d)$

40. To maintain transactional integrity and database consistency DBMS will use:

(A) Triggers
(B) Cursors
(C) Locks
(D) Pointers

41. The design of a class in C++ which can’t be inherited is achieved using ___ keyword before class declaration.

(A) Final
(B) Static
(C) Sealed
(D) Fixed

42. CPU register to perform storage of arithmetic and logic results is called:

(A) Instruction register
(B) Program counter
(C) Accumulator
(D) Instruction Decoder

43.The nature of any number i.e., positive or negative is recognized by its:

(A) MSB
(B) LSB
(C) Bits
(D) Nibble

44. The output of the following code is:

```
int main()
{
   static int x[ ]={1,2,3,4,5,6,7,8}
   int i;
   for(i=2;i<6;++i)
      x[x[i]]=x[i];
   for(i=0;i<8;++i)
      printf(“%d”,x[i]);
   Return 0;
}
```

(A) 12244668
(B) 11335578
(C) 12335578

(D) 12343676

45. Using Mid Point algorithm, the new coordinates of the point P(x,y) having slope ‘m’ and constant and ‘b’ is calculated using:

(A) F(x,y)=mx+b-y
(B) F(x,y)=mx-y+b
(C) F(x,y)=mx+b
(D) F(x,y)=mx-y+b-y

46. The theory of bubbled input OR gate is interchangeable with a bubbled output AND gate is demonstrated and proved by:

(A) Karnaugh map
(B) DeMorgan’s second theorem
(C) The commutative law of addition
(D) The associative law of multiplication

47. A DFD that can be used to plan or record the specific makeup of a system is called:

(A) Level 0 DFD
(B) Level 1 DFD
(C) Level 2 DFD
(D) Level 3 DFD

48. For Providing large storage space the semiconductor based storage is not permitted now a days due to their:

(A) Lack of sufficient resources
(B) High cost per bit value
(C) Lack of speed of operation
(D) High cost of raw material

49. Let the next generation computer systems of 64 bit has virtual address space is of the same size as the physical address space. In such a case, if we remove the virtual space completely in new systems then which one of the following is true in this scenario?

(A) Efficient implementation of multi user supports is no longer possible
(B) The processor cache organization can be made more efficient now
(C) Hardware support for memory management is no longer needed
(D) CPU scheduling can be made more efficient now

50. Which of the following statements is false?

A. There exist parsing algorithms for some programming languages whose complexities are less than $O(n^3)$

B. A programming language which allows recursion can be implemented with static storage allocation

C. L-attributed definition can be evaluated in the framework of bottom-up parsing

D. Code improving transformation can be performed at both source language and intermediate code level.

51. Which of the following is FALSE about inheritance in Java?
(A) Private methods are final
(B) Protected members are accessible within a package and inherited classes outside the package
(C) Protected methods are final
(D) We cannot override private methods

52. Given a mask, M=255.255.255.248. How many subnet bits are required for given mask M?
(A) 2
(B) 3
(C) 4
(D) 5

53. Transparent mode firewall is a:
(A) Layer 2 firewall
(B) Layer 3 firewall
(C) Layer 4 firewall
(D) Layer 7 firewall

54. The situation when we can't use a SAX parser is:
(A) We can process the XML document in a linear fashion from the top to down
(B) We are processing a very large XML document whose DOM tree would consume too much memory
(C) The problem to be solved involves only part of the XML document
(D) When we are parsing HTML documents to read and store the contents written in fields.

55. A routing protocol that was used for trail information that gets updated dynamically is:
(A) Distance Vector
(B) Path Vector
(C) Link Vector
(D) Multipoint

56. The catalan number for generating different binary trees is given by:
(A) !2n / (!n*!(n+1))
(B) C(n,2n) (n+1)
(C) !n*C(2n,n) (n+1)
(D) C(2n,n)/ !n

57. Which of the following problems are decidable?
a. Does a given ever produce an output?
b. If L is a context free language, then is L' (complement of L) also context free?
c. If L is a regular language, then is L' also regular?
d. If L is a recursive language, then is L' also recursive?
(A) (a), (b), (c), (d)
(B) (a), (b)
(C) (b), (c), (d)

(D) (c), (d)

58. If G be an undirected connected graphs with distinct edge weight such that maximum weight and minimum weight of edge is given by emax and emin respectively. Then, Which of the following statements is TRUE?

(A) Every minimum spanning tree of G must contain e_{min} and e_{max}
(B) If e_{max} is in a minimum spanning tree, then its removal must disconnect G
(C) No minimum spanning tree contains e_{max}
(D) G has a multiple minimum spanning trees

59. Consider bottom up merge sort working on 'n' elements such that $n=2^i$. The minimum number of comparisons in order to get sorted list is:

(A) nlog(n/2)
(B) nlogn
(C) (nlogn)/2
(D) logn

60. Which among the following can be solved with master theorem?

(A) T(n/2)+n(2-cosn)
(B) $64T(n/8)-n^2logn$
(C) 16T(n/4)+n!
(D) 0.5T(n/2) + 1/n

61. The best example of compile time polymorphism is:

(A) Method declaration
(B) Method overriding
(C) Method overloading
(D) Method Invoking

62. The correct order of the degrees of vertices that would form the connected graph as eulerian is:

(A) 1,2,3
(B) 2,3,4
(C) 2,4,5
(D) 1,3,5

63. Consider a B+ tree in which the maximum number of child nodes is 6. What is the minimum and maximum number of keys in such a tree?

(A) 3 and 4
(B) 2 and 5
(C) 2 and 4
(D) 3 and 5

64. The primary key of any table is selected from the:

(A) Composite keys
(B) Foreign keys

(C) Candidate keys
(D) Alternate keys

65. Which of these classes is not part of Java’s collection framework?
(A) Map
(B) Array
(C) Arraylist
(D) Vector

66. The time complexity of the Tarjan’s algorithm for finding the strongly connected components of a graph having m vertices and n edges is:
(A) O(m*n)
(B) O(m+n)
(C) $O(m^2)$
(D) $O(m^n)$

67. A planar graph is defined as:
(A) A graph drawn in a plane in such a way that any pair of edges meet only at their end vertices
(B) A graph drawn in a plane in such a way that if the vertex set of graph can be partitioned into two non-empty disjoint subset X and Y in such a way that each edge of G has one end in X and one end in Y.
(C) A simple graph which is Isomorphism to Hamiltonian graph.
(D) A function between the vertex sets of two graphs that maps adjacent vertices to adjacent vertices

68. The process of rotating the image by 180^0 without changing its size is called:
(A) Translation
(B) Rotation
(C) Scaling
(D) Reflection

69. Which IC is used for the implementation of 1 to 16 DEMUX?
(A) IC 74154
(B) IC 74155
(C) IC 74139
(D) IC 74138

70. Which of the following statements is true?
(A) If a language is context free it can always be accepted by a deterministic pushdown automaton
(B) The union of two context free language is context free
(C) The intersection of two context free language is context free
(D) The complement of a context free language is context free

71. Frequency hopping spread spectrum can be utilized to solve the problem of:
(A) Data transfer
(B) Enhancing signal strength
(C) Interference between signals
(D) Connection establishment

72. Given the sequence of nodes {11,6,8,19,4,10,5,17,43,49,31} not necessarily in correct order for generating binary search tree. The corresponding postorder traversal of the BST is:
(A) 5,4,10,8,6,49,31,43,19,17,11
(B) 4,5,6,8,10,11,19,17,43,31,49
(C) 4,5,8,10,6,17,31,49,43,19,11
(D) 5,4,10,8,6,17,31,49,43,19,11

73. A lock which required is not acquired by database engine is:
(A) Row lock
(B) Page lock
(C) Table lock
(D) Attribute lock

74. A complete binary tree with the property $K_{ni}>=K_{cn}$ where K_n is n^{th} node value and K_c be the value of its child node is called:
(A) B+ tree
(B) Threaded binary tree
(C) Heap
(D) AVL tree

75. The name of parser generator that is used for SQL query parsing is:
(A) Lexer parser generator
(B) Syntactic parser generator
(C) Tokenizer parser generator
(D) SQL DML parser generator

76. In the Merge sort algorithm, the time taken by the algorithm to divide an array of size n into two equal halves and many times that divide is called in overall sorting procedure is:
(A) O(1) and Θ(logn)
(B) Θ(1) and Θ(logn
(C) O(logn) and Θ(logn)
(D) Θ(logn) and O(1)

77. The best example of language for defining the non regular languages is:
(A) Languages for palindrome and prime
(B) Languages for palindrome and Even-Even
(C) Language for prime and Even-Even
(D) Language for palindrome and factorial

78.Consider an undirected random graph of 16 vertices. The probability that there is an edge between a pair of vertices is ½. What is the expected number of unordered cycles of length three?
(A) 70
(B) 280
(C) 120
(D) 45

79. ___ is a parameter passing method that waits to evaluate the parameter value until it is used?
(A) Pass by value
(B) Pass by name
(C) Pass by reference
(D) Pass by pointer

80. Suppose that we have an MxN array ‘A’ in row major order, where the size of the elements is S. Then the location of A[i,j] elements is given by:
(A) A+i*N*S
(B) A+(i*N+j)*S
(C) A+(i*N+j*M)*S
(D) A+(i*M+j*N)*S

81. A friend function in C++ can’t be used with:
(A) Member function
(B) Object
(C) Class
(D) Class template

82. The quantity of very long word is:
(A) 32 bits
(B) 64 bits
(C) 128 bits
(D) 256 bits

83. Given language L={$a^{(2*m)}c^{(4*n)}d^n b^m$ | m,n>=0}. Find the best appropriate match to recognize L.
(A) Finite automata
(B) Pushdown automata
(C) Non deterministic turing machine
(D) Non deterministic PDA

84. Suppose the pipelined stages take {15,8,25,4,8} nanoseconds(ns) respectively. With a buffering delay of 5ns. Two pipelined implementation of the processor are used?
(i) A naive pipeline implementation(NP) and
(ii) An efficient pipeline(EP) where the third stage is splitted into {12ns, 5ns and 8ns} respectively.

The speedup achieved by EP over NP in executing 100 independent instructions with no hazards is:

(A) 1.471
(B) 1.517
(C) 1.638
(D) 1.567

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85. Consider the problem of determining string ‘w’ is given some turing machine ‘M’ will it enter into some state ‘q’, where q belongs to set of all states in Turing machine M and sting w is not equal to empty string. Which among the following is correct for above statement?

(A) Given problems is computable
(B) Given problem is non computable
(C) Given problem has finite state solution
(D) Given problem is solved in polynomial time.